

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Saturday, 12 September 2026, 9:46 PM**State** Finished**Completed on** Saturday, 12 September 2026, 9:47 PM**Time taken** 50 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5	12 34 67 78 98
67 34 12 98 78	

Answer:

```

1  #include <stdio.h>
2
3  void swap(int *a, int *b)
4  {
5      int temp = *a;
6      *a = *b;
7      *b = temp;
8  }
9
10 int partition(int arr[], int low, int high)
11 {
12     int pivot = arr[high];
13     int i = low - 1;
14
15     for (int j = low; j < high; j++)
16     {
17         if (arr[j] <= pivot)
18         {
19             i++;
20             swap(&arr[i], &arr[j]);
21         }
22     }
23
24     swap(&arr[i + 1], &arr[high]);
25
26     return i + 1;
27 }
28
29 void quickSort(int arr[], int low, int high)
30 {
31     if (low < high)
32     {
33         int p = partition(arr, low, high);
34
35         quickSort(arr, low, p - 1);
36         quickSort(arr, p + 1, high);
37     }
38 }
39
40 int main()
41 {
42     int n;
43
44     scanf("%d", &n);
45
46     int arr[n];
47
48     for (int i = 0; i < n; i++)
49         scanf("%d", &arr[i]);
50
51     quickSort(arr, 0, n - 1);
52

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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